Low-Cost Carriers and the Transformation of the Indian Domestic Aviation Sector

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ABSTRACT

This paper attempts to describe the growth and impact of low-cost carriers (LCCs) in India's domestic aviation industry. It charts how the LCC model grew in tandem with the evolution of airline involvement, domestic connections, passenger usage, and operational, safety, and pricing systems. Secondary data gathered in this study include information from aviation authorities, airline disclosures, published data sets, flight tracking data, and some academic and industry literature. The analysis span is 2000–2024 with selected observations from 2025 to note the recent operational and regulatory developments. Instead of making causal statements, the current paper groups the evidence descriptively and highlights the key features of the Indian LCC model. The results reveal a sector that has been affected by competitive low operating costs, network expansion, shifting market participation, and network model variations alongside the ongoing regulatory oversight. The paper also documents the challenges that may occur in a highly-used airline system when it has staffing, scheduling or regulatory requirements. The final report sheds light on the development of LCCs in the context of the overall structural change of domestic air transport in India.

Keywords: low-cost carriers; Indian aviation; domestic air transport; competition; connectivity; operations; safety; fares; aviation regulation

1. INTRODUCTION

Domestic aviation sector has experienced significant change in the structure of the industry since the liberalization of the sector and emergence of low-cost airlines in India. Air Deccan is credited with introducing LCC model at an early stage of development and IndiGo is a good example of the fleet standardization and discipline and standardized growth with size. Other carriers such as SpiceJet, AirAsia India, Vistara, TruJet and Akasa Air joined the fray at various points. The industry also saw the closure of airlines, mergers, takeovers and major disruptions due to the global financial crisis and the COVID-19 pandemic. These changes justify the use of a longitudinal descriptive method for a description of the changes in the composition and operating characteristics of the market over time. The current paper therefore analyzes the development of LCC in five descriptive aspects – market structure, accessibility and connectivity, operational practices, safety and regulatory context and fare and service structure.

2. LITERATURE REVIEW

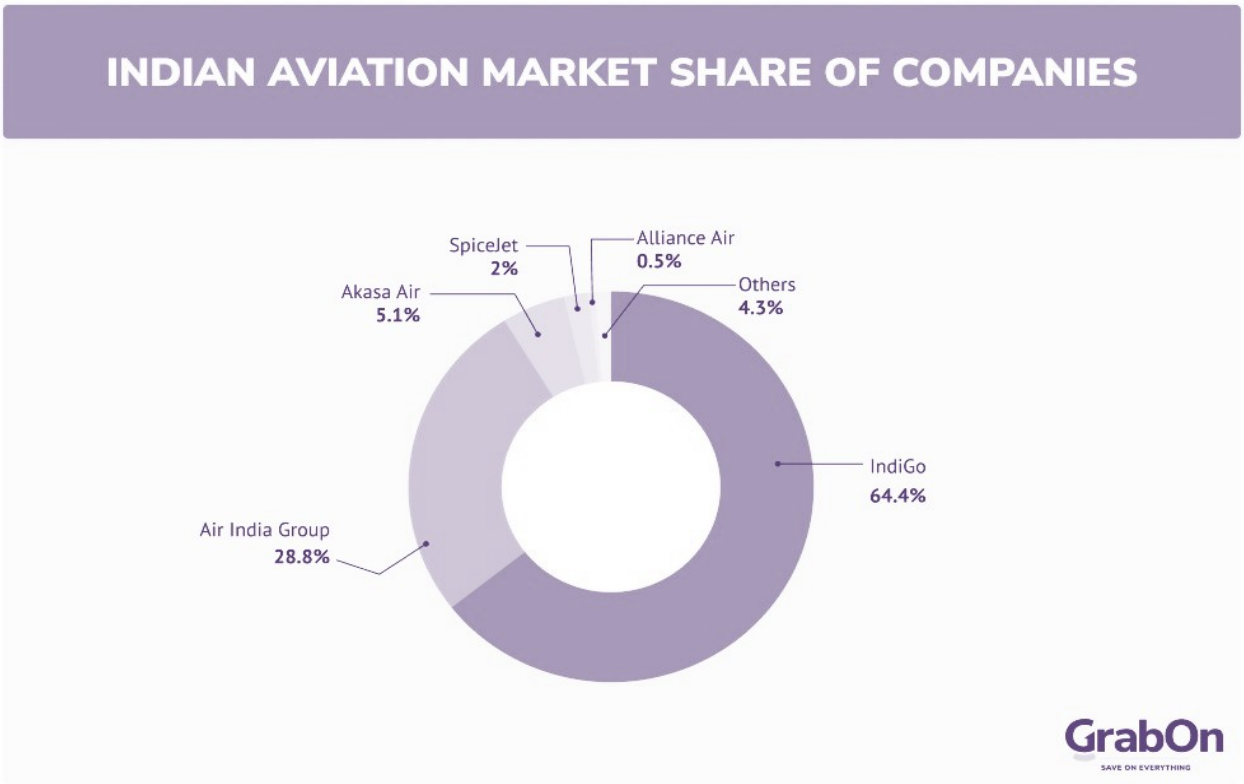
2.1 LCCs and Market Competition

The literature on LCCs presents a common set of features that are linked to the LCC model: simplification of service, unbundling of fares, relatively standardized fleets, high aircraft utilization, fast aircraft turnaround and cost discipline-oriented operating structure (Toni et al., 2025). The birth of Air Deccan in the early 2000s and the subsequent development of the domestic model of low-cost carriers

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(LCC), IndiGo, is often mentioned in connection with the development of the domestic LCC model. Fleet commonality and fleet scale are also important characteristics of the model and are explored in the literature, as are the ways in which carriers respond to market changes in their networks and service offerings (Gopinath, 2005).

Figure 1. Market share of Indian-operated carriers, FY 2025. Source: Shewale (2026a), Indian Aviation Statistics: Complete Market Analysis.



2.2 Accessibility and Connectivity

In the literature, accessibility is related closely to the fare levels, availability of routes, flight frequency, and geographical distribution of airports. LCC services in India grew, providing alternative air-travel choices for passengers sensitive to the cost of air travel, and helped establish linkages outside the largest metropolitan markets. Concurrently, the literature warns that the headline price is not the full cost of travelling when ancillary products such as food and seat selection, priority services and/or baggage are sold separately. Consumer access is thus best understood as both the basic fare and the overall service offering. This broader development includes regional-connectivity programmes like UDAN.

2.3 Operational Efficiency and Innovation

Efficiency is one of the core elements of LCC business model. The published literature is linked to standardisation of processes, fleet commonality, high aircraft utilisation, quick turnaround, and disciplined cost management (Toni et al., 2025). IndiGo is widely talked about in India due to its focus on a fairly uniform fleet and high utilization. These characteristics are not necessarily indicators of a single cause and effect, but rather of practices put in place when combined with maintenance, staffing, scheduling, and regulatory requirements.

2.4 Safety and Regulatory Compliance

In the aviation industry, safety is regulated in various aspects including airworthiness, flight operations, crew licensing, and safety management systems, with the regulatory framework being provided in India by the Airports Authority of India. In Indian civil aviation, safety is regulated under various aspects such as airworthiness, flight operations, crew licensing and safety management systems under the regulatory framework provided by the Airports Authority of India. ICAO standards set an international framework and the Directorate General of Civil Aviation (DGCA) oversees the regulations in India (International Civil Aviation Organization, 2020). The LCC model is not considered a separate safety regime in the literature but rather an approach. Rather, the safety requirements are applied to all carriers, and the differences in fleet structure, maintenance structures, staffing, and utilization and operation complexity could impact the way individual airlines perform the application. Financial and supply-chain pressures can also impact aircraft availability and maintenance planning, which are part of the safety context of the industry, and may impact regulatory compliance and operational resilience.

2.5 Identified Research Gaps.

The literature review suggests that there are significant opportunities for route-level and market-level descriptive research on the evolving Indian aviation structure. Past studies have covered aspects of pricing, competition, service quality, airline operational models and airline development, and recent events like the airline consolidation process, the downfall of Jet Airways, the bankruptcy of Go First, and the growth of newer air carriers have affected the market under study. This paper fills this gap in description by combining market, connectivity, operational, safety and fare data into one longitudinal description.

2.6 Historical Development of the Indian LCC Model

This time also is a time of significant disruption. World economic turmoil influenced aviation markets and the COVID-19 pandemic created an unprecedented disruption to domestic and international aviation travel. These are the airline exit and participation changes during and after the periods mentioned that make up the historical record examined here. However, the paper does not include temporary disruptions as illustrations of a general underlying relationship if the paper treats such events as contextual features.

The Indian experience also shows the interactions between the airline strategy and the broader transport system. There was a greater focus on the LCC model, emphasising low base fares, operational ease of services and aircraft, and a more pro-active approach from public policy towards regional connectivity. This resulted in a network of high volume routes and routes to smaller cities. This is seen in the airport and the passenger-activity series to follow in this paper. The expansion of airports or passenger activity cannot be wholly explained by LCCs as it has been observed that airport investment, household income, tourism, business travel, regulation and wider economic conditions also affect air-transport demand.

The sector's history is not a straightforward model change, from one business model to another. Multiple carriers came and went with various commercial models and some changed ownership, service offerings, or exited the market. The carrier-presence table presented in this paper, then, is not just a record of the number of airlines listed at each point; it has descriptive value as well. It shows that the market has been different in different periods and that some airlines have been playing for a while. It is significant that some of the carriers included are no longer in service; otherwise, a material part of the development of the sector would be lost.

It is important to recognize that liberalization of India's domestic airline industry is part of the development of low-cost aviation in India. As indicated in the literature provided, the modern LCC model started to emerge in the early 2000s when the market began to be dominated by private participation and new modes of airline organization. In the literature, Air Deccan is frequently referred to as an early and significant instance of the model, with IndiGo's more recent introduction highlighted as a more developed form of low-cost airline, based on fleet commonality, process standardization and scale.

3. STUDY AIM AND SCOPE

3.1 Aim of the Descriptive Study

The aim of the study is to document the development of the low-cost-carrier model within India's domestic aviation sector and to describe the principal changes associated with its expansion. The paper focuses on observable characteristics of the sector rather than attempting to establish causal relationships between LCC entry and individual market outcomes.

The descriptive framework brings together four connected areas of observation: changes in the population of operating carriers; changes in connectivity and passenger activity; operational characteristics associated with LCCs; and the development of fare, service, safety and regulatory arrangements.



Figure 2. Descriptive framework for examining the development of low-cost carriers in the Indian domestic aviation sector. Source: Author.

The framework views the development of LCCs as more than just another trend in the organisation of domestic air transport, it is a part of a greater change. Market entry, airline participation changes, network

development, operating practices and fare structures are all bundled together as each represents a different aspect of the sector's development.

The study doesn't attribute a causal effect to any one factor. Rather it documents the relationships and changes that can be discerned in the existing second-hand resources and puts them into perspective of the regulatory environment, economic trends, airline failures, and broader shifts in the Indian air transport arena.

3.2 Descriptive Characteristics of the LCC Business Model

The model should also differ from a formula of what an airline can and cannot offer. Airlines can operate with low costs and provide varying levels of service, network size and product variety. The historical literature mentioned in the manuscript outlines the evolution of the semi-premium and diversified LCC models elsewhere, and Indian carriers have also evolved their networks and products over time. The descriptive nature of this study is therefore to define some common features of the model in the Indian evidence, rather than to type categorize all airlines in one defined format.

The LCC model also distinguishes the base-level charges from a series of optional charges. Products such as baggage, seat selection, food and beverages and priority services can be charged separately. This means that the cost that is advertised and the cost that the passenger pays and pays for may not be the same. A description of the fares should therefore specify what the data underlying the fares measures. The observations made by Goibibo were kept in this paper as fare observations and not as a proxy for the total cost of trip to and from the station for all passengers.

The relevance of fleet structure is more important in the descriptive analysis as commonality in the fleet can help in reducing the number of aircraft types that an airline has to support in training, maintenance and operational planning. Not all of the carriers discussed in this paper are necessarily of the same fleet configuration. The aircraft information provided for Air India is significantly more varied than for Akasa Air and IndiGo is fairly evenly distributed across aircraft families, which are mostly related to narrow-body domestic services. These differences should be taken into account when making observations and interpreting operating-cost indicators when comparing turnarounds, but they do not imply that one fleet configuration is better than another.

In the literature the low cost carrier model is defined as more than a pricing concept, it is a coordinated operating system. Its main features are simplified service, the separation of optional services from the basic ticket, the relatively standardised fleets, high aircraft utilisation, and ground turnaround times and procedures that are geared toward minimising unnecessary complexity. These are all related: the idea of a ‘simplified' product is not just one of lowering prices, but also one of providing an airline with a way to operate aircraft, staff and resources predictably.

4. METHODOLOGY

4.1 Research Design

The study is descriptive in nature and adopts a longitudinal research design, completely relying on secondary data related to the domestic aviation sector of India. The time period of observation is 2000-2024, with selected observations for 2025 as appropriate for the development of current operational or

regulatory issues. The design should document change across time and between carrier types and is not designed to estimate causal effects nor to test statistical hypotheses.

4.2 Data Sources

The principal sources used for the description are the Directorate General of Civil Aviation (DGCA), published airline data, Flightradar24 flight records, Statbase fleet records, the cited data from the dataset of Goibibo, aviation-grievance data, published market-share data, and selected academic and institutional literature. These sources are only used for the domains of description that are included in the retained figures and text of the revised paper. Sources from the original manuscript that are not traceable back to a retained datum have not been used as evidence for a particular finding. Published secondary figures are differentiated from author-created and author-derived figures in the reference list and data-provenance register.

4.3 Population and Temporal Scope

Indian domestic aviation market is the population of interest. The study is not a sample study nor a survey, but instead a study of secondary records of airlines, routes, airports, passengers, flights, fares, operational characteristics and reported aviation incidents. The main longitudinal period runs from 2000 to 2024, with some observations from 2025 retained only when they provide context for developments from the current operating and regulatory context.

The collection of data and matching of sources. The gathering of data and matching of source material.

The sources identified in the reference list were used to compile data which were then sorted in relation to the descriptive topic to which each record is related. The observations on market structure were matched to aviation-authority data, passenger and airport observations to the cited aviation data, operational observations to flight-tracking and fleet data, incident observations to the cited aviation-grievance data, and fare and flight-volume observations to the cited Goibibo data. If the figure is one to which the manuscript attributes the creation of source, or to the calculation of source from other figures, the location of the figure in the source where it was created or calculated and the type of transformation are documented in the data-provenance table.

4.5 Data Domains

The airline participation and market structure domain, domestic passenger activity, airport and regional connectivity, airline network coverage, aircraft fleet characteristics, aircraft turnaround, operating-cost indicators, reported incidents, domestic air fares, and flight activity domains make up the descriptive dataset. The study also documents relevant context on liberalisation, airline exits, consolidation, COVID-19 disruption and regulatory developments. In the operating-cost material, CASK is used as the abbreviation for cost per available seat kilometre.

Information about operational costs is used as a descriptive indicator of the characteristics of the airline's operations. If a figure was compiled from values from several sources, the figure caption and the provenance register refer to the source material that was available in the manuscript. The new mathematical relation is not implied in the updated paper; the calculated indicator are only reported as descriptive values in the retained figures.

Data preparation and descriptive treatment, 4.6.

The source material was reviewed, organized and compared using Microsoft Excel. Records were aggregated to each of the following: id by year, airline, route or descriptive category, based on the dataset. In the original study, it was decided to exclude itineraries that included more than one stop and keep in the study only those itineraries that were direct flights, for the purpose of fare observation. These observations are descriptively reported in the revised study and do not involve inferential statistical testing or causal estimation but only descriptive comparison. Examples of graphs are not used as statistical tests but as visual descriptions of the data sets.

4.6.1 Source Hierarchy and Data Verification.

The verification process involved verifying that the description of a figure matched the source of the figure, that the dates and the carrier names were internally consistent and that the interpretations went beyond what could be reasonably concluded on the basis of the observation. The numerical observations that are included within the graphs that were retained remain part of the source-based visual record, but do not engage in any inferential statistical testing or set causal effects from the revised prose.

In cases where the original source did not give an adequate definition of a derived value, the paper now explains the value as it was derived through the procedure indicated in the original manuscript provided. The turnaround figure is included as an observation made by the author based on common route and recorded aircraft movements. It is not given as a company-wide average. The same holds true for the CASK comparison, which is presented as an author-derived operating-cost comparison, and whose source limitations are explicitly specified in the provenance register.

The Sources were identified for each figure and given priority. For the incident data, flight-tracking data was obtained with DGCA data and financial data retained for the basis of the operating-cost observation and Statbase data for the fleet-age data, while the Flightradar24 data was used for flight-tracking and Kaggle data for financial data and observations on flight volumes. The paper does not use other datasets that are selected specifically to achieve uniformity of figures.

Since the paper makes use of multiple sources of secondary evidence, source matching is not treated as an after-thought of editing, but rather as a method. Each figure or table has been referenced to the source given in the manuscript, and the provenance register indicates whether the material was reproduced verbatim or had been collated by the authors or is based on an observation or calculation. This is significant because an author-provided graph may evidence a different meaning than a published series, even if both are rendered in the same visual style.

4.7 Ethical Considerations

The study does not involve human participants, interviews or the collection of personally identifiable information but solely relies on secondary information. Other data that is openly available, published documents and recorded aviation data are credited to their sources. The data-provenance register is incorporated to make it clear where and how the main figures have been treated.

4.7.1 Data Classification and Quality Controls

Quality control was based on source tracing, internal consistency and methodological transparency. The authors checked whether each figure could be connected to an identified source and whether the accompanying prose accurately represented the nature of the data. Where the available record was

incomplete, the limitation is stated rather than concealed. This is particularly important for incident records and author-derived operating measures, for which differences in reporting practices can affect comparability between carriers.

Temporal matching was also considered during data preparation. The principal descriptive period is 2000–2024, while selected observations from 2025 are included where the manuscript specifically addresses recent operational or regulatory developments. These observations are not presented as if they belonged to the same historical series without qualification. Where a graph covers a narrower period, its caption identifies that period and its associated source. This allows the reader to distinguish long-run historical description from recent case evidence.

Carrier classification requires particular care because airline business models can change over time. An airline identified as a low-cost carrier in one period may later change its product, ownership, network structure or relationship with another airline group. The study therefore follows the classifications and carrier names contained in the supplied source material rather than retrospectively imposing a single modern classification on the entire historical period. This approach is especially relevant to the historical carrier table, which includes airlines that later ceased operations or changed status.

The data were classified according to the type of phenomenon being described. Market-structure records identify carriers participating in the domestic market; passenger records describe traffic activity; airport records describe infrastructure and connectivity; flight-tracking observations describe aircraft movement between an arrival and a subsequent departure; fleet records describe the age and composition of aircraft; fare records describe ticket-price observations; and incident records describe reported events contained in the cited dataset. Keeping these domains separate reduces the risk of treating unlike observations as if they were measurements of the same construct.

4.8 Methodological Limitations

The descriptive record can only be limited by the classification of airlines, definitions of sources, lack of data in the past, underreporting of incidents, changes in the structure of the market and influence of major external events from foreign airlines. For some, it is author-created datasets or author-derived observations due to the lack of an official series that is directly comparable to the data. These restrictions place limitations on what conclusions this paper draws and preclude an exhaustive explanation of the results of all changes observed as being the product of LCC activity.

5. DESCRIPTIVE FINDINGS

5.1 Market Structure and Carrier Participation

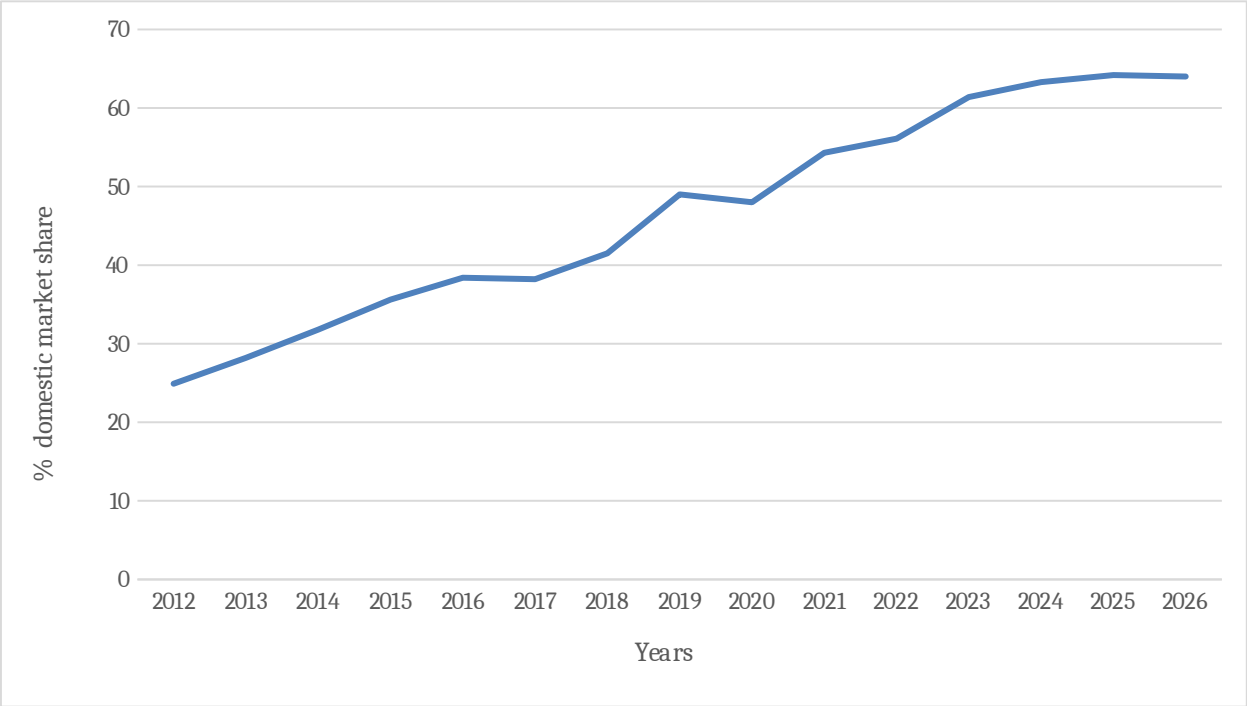
The emergence of Low-Cost Carriers is part of a larger shift in the domestic aviation industry in India. After liberalisation the market consisted of varying mix of full-service and low-cost players. The LCC model was based on lower base fares, using a simplified service, common aircraft fleet, high utilisation and fast turnaround. The incumbent and full-service carriers then competed with the LCCs and a few carriers came and went or evolved their services. These changes are documented by the retained figures which show the market-share and carrier-participation records.

Table 2. Data provenance and source register for the principal figures and datasets. Table made by author.

Data / Figure	Data represented	Source identified in manuscript	Treatment / Provenance
Figure 1	Indian market share of carriers operated by Indians	Shewale (2026a), *Indian Aviation Statistics: Complete Market Analysis* (GrabOn)	Published secondary data reproduced in the figure.
Figure 2	Conceptual/descriptive framework	Author-developed	Author-created framework; no external dataset used.
Figure 3	IndiGo domestic market-share progression	Global Indians LinkedIn post (2024)	Published secondary data reproduced in the figure.
Figure 4	Number of airlines in the Indian domestic market	DGCA	Author-created competition dataset organised from DGCA data.
Table 1	Airlines present at selected four-year intervals	DGCA	Author-adapted tabulation of carrier presence using DGCA data.
Figure 6	Number of airports in India by year	Indboard	Published secondary data reproduced in the figure.
Figure 7	Domestic passengers by year	Directorate General of Civil Aviation (DGCA), Government of India	Published DGCA data reproduced in the figure.
Figure 8	Average turnaround time by airline	Flightradar24	Author-derived observations from landing and departure data on a common route. No official comparable turnaround-time series was identified.
Figure 9	Average fleet age by airline	Statbase, *Aircraft Fleet Average Age by Airlines*	Published secondary data adopted for the figure.
Figure 10	Fleet age and CASK by airline	Times of India; airline operating-cost and capacity data identified in the manuscript	Fleet-age figures were compared with CASK. CASK values for other carriers were calculated by the author from operating cost, seat-capacity and distance data. The supplied manuscript does not identify the individual source document for each carrier.
Figure 11	Reported incidents by airline	Tiwari (2022), *India Aviation Grievance Dataset*, Kaggle	Reported incident totals reproduced from the dataset. These figures are not treated as a direct measure of aviation safety.
Figure 12	Domestic fare observations by carrier, 2023	Sahu (2024), *Goibibo Flight Data*, Kaggle (Kaggle, 2023)	Secondary fare observations reproduced from the dataset.
Figure 13	Flight volumes by airline, 2019–2025	Sahu (2024), *Goibibo Flight Data*, Kaggle	Flight-volume observations reproduced from the dataset.

Source-transparency note. The supplied manuscript does not individually identify the official carrier documents underlying every operating-cost value used in Figure 10. Rather than infer or substitute sources, this revised paper records that limitation explicitly. All other principal figures are linked to the source or author-created dataset identified in the manuscript's reference list.

Figure 3. IndiGo domestic market-share progression. Data Source: Global Indians (2024), LinkedIn. Graph made by the author.



Since the arrival of IndiGo, it became a main player in the domestic market and is a frequently cited case in the literature and source material for an LCC with a fleet standardization, scale and high utilization operating model. The material also captures the growth of airspace newcomers like Akasa Air in a rapidly shifting market that is consolidating. The carrier scenario hasn't been unchanged. Coinciding with the arrival of airlines have also come airline exits, ownership changes, and consolidations. This is why the description of the market does not only include the airlines that are still operating, and their presence is kept in the longitudinal record even those that have ceased operating.

The carrier series shows changes in the composition of the domestic market across the study period. The accompanying record should be read as a description of market participation rather than as evidence that LCC entry alone caused changes in the number of carriers.

The table identifies the carriers recorded in the supplied dataset at selected points in the study period. It illustrates the entry of new operators, the disappearance of earlier operators and the changing mixture of LCC and full-service business models. Because the table is based on selected years, it is intended as a structural snapshot rather than a complete year-by-year register.

Airline exits are an important part of the descriptive record. They demonstrate that participation in the Indian domestic market has changed repeatedly in response to commercial conditions, industry consolidation and wider disruptions.

Figure 4. Number of airlines represented in the Indian domestic market over the study period. Source: author-created competition dataset based on data from DGCA – 2000-2024.

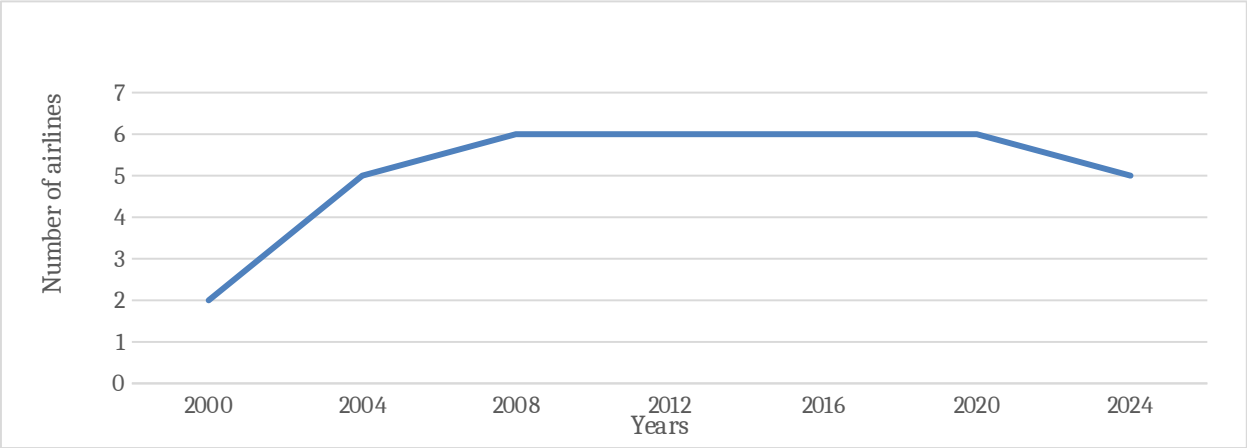


Table 1. Airlines represented in the Indian domestic market at selected four-year intervals. Source: author-created competition dataset based on DGCA data.

2000					
Indian Airlines	Air India				
2004					
Indian Airlines	Air India	Air Deccan	Air Sahara	Kingfisher Airlines	
2008					
Air India	SpiceJet	GoAir	IndiGo	JetLite	Kingfisher Airlines
2012					
Air India	SpiceJet	GoAir	IndiGo	JetLite	Kingfisher Airlines
2016					
Air India	SpiceJet	GoAir	IndiGo	TruJet	Vistara
2020					
IndiGo	Air India	AirAsia India	Akasa Air	TruJet	Vistara
2024					
IndiGo	Air India	AirAsia India	Akasa Air	Vistara	

5.2 Connectivity, Airports and Passenger Activity

The development of domestic aviation has also involved changes in the geographical distribution of air connectivity. The supplied material describes the expansion of airport activity beyond the largest metropolitan centers and identifies regional-connectivity initiatives, including UDAN, as part of this development. LCC networks are described as particularly relevant to connections involving Tier 2 and Tier 3 cities.

The domestic market includes both major metropolitan airports and regional airports serving smaller cities. The supplied material describes IndiGo, SpiceJet and Air India as serving regional markets to different extents, with LCCs maintaining a notable presence in Tier 2 and Tier 3 connectivity. The extent and composition of these networks have changed as airlines have entered, withdrawn from and reorganised routes.

Figure 6. Number of airports in India by year. Data Source: Indboard, Number of Airports in India – Annual Growth, 2004–2024. Graph made by author.

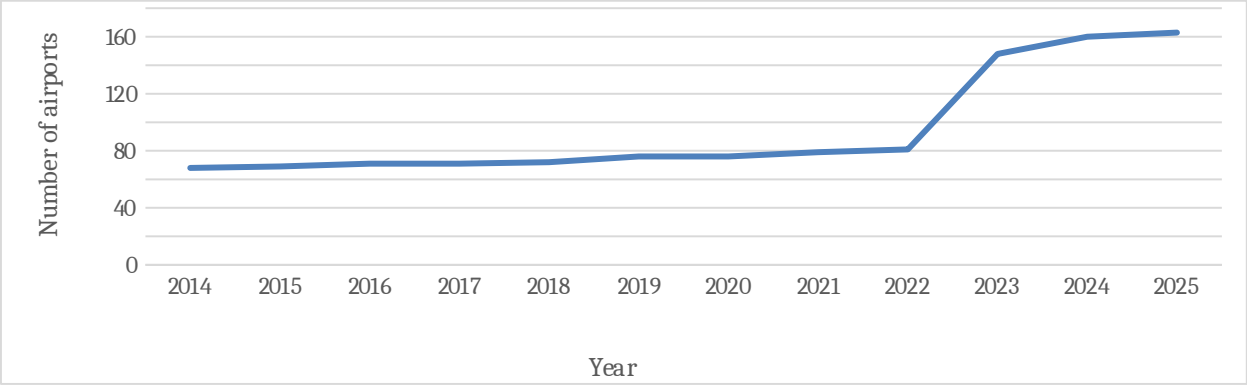
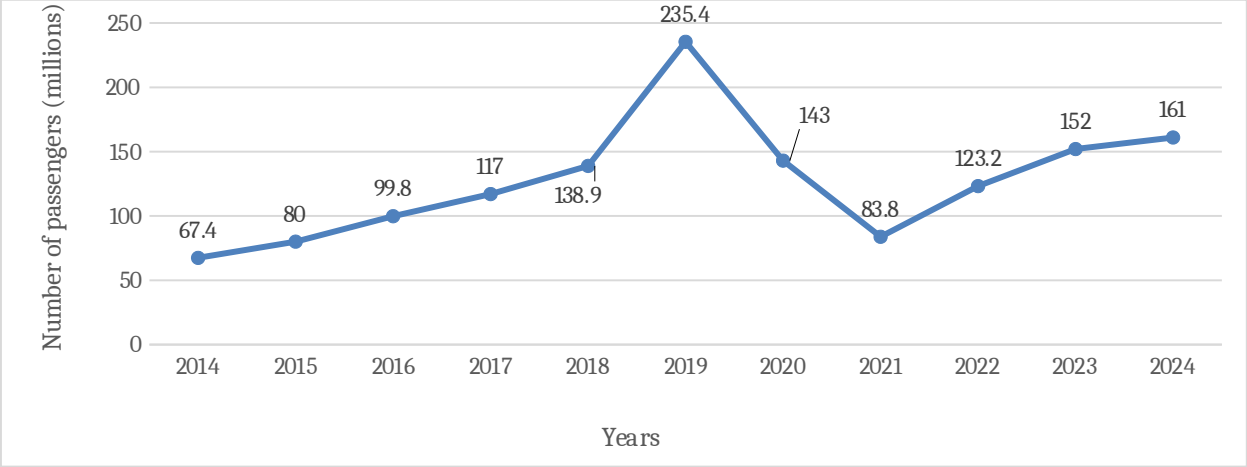


Figure 7. Domestic passenger activity by year. Data Source: Directorate General of Civil Aviation (DGCA), Government of India. Graph made by author.



IndiGo

The supplied research record describes IndiGo's domestic network as extending beyond the largest metropolitan markets into Tier 2 and Tier 3 destinations. The carrier's participation in the UDAN regional-connectivity programme is also recorded in the source material.

SpiceJet

The source material describes SpiceJet as operating a network that includes regional and smaller-city destinations, including destinations covered by the UDAN programme. Its network is presented as extending across several regions of India.

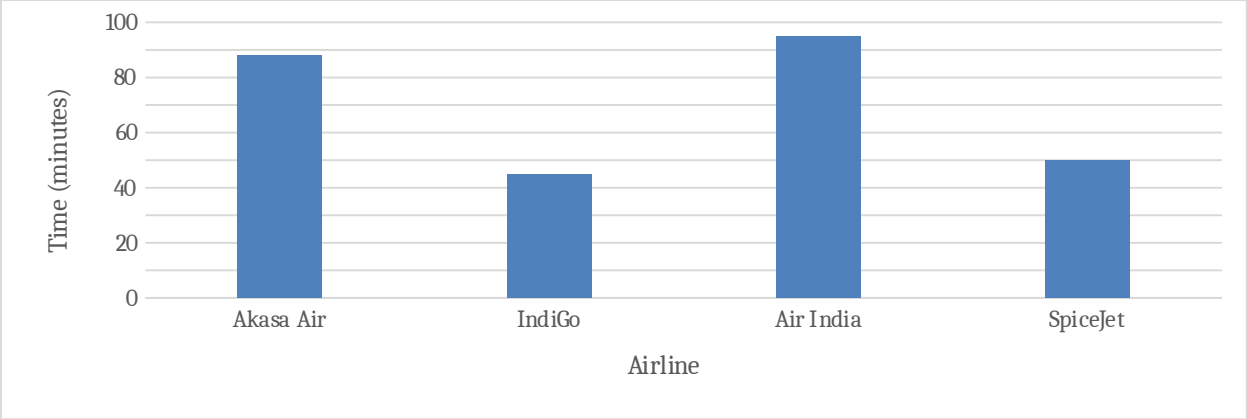
Air India

The supplied material describes Air India's domestic network as serving metropolitan centres, state capitals and selected Tier 2 and Tier 3 destinations. It also distinguishes Air India's regional presence from the broader regional coverage associated with LCC networks.

5.3 Operational Characteristics and Fleet Structure

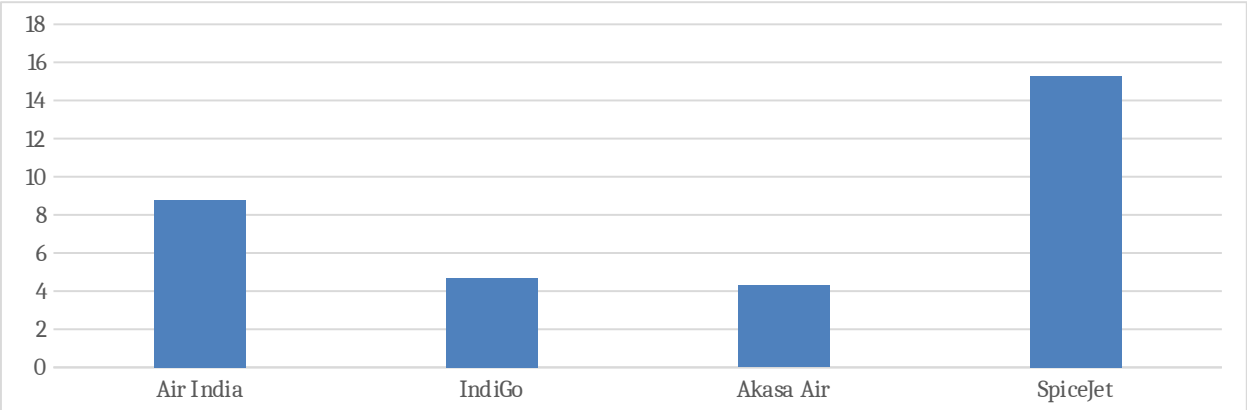
The LCC model is associated in the literature and source material with operational practices intended to simplify and accelerate aircraft operations. The retained figures examine turnaround, fleet age and operating-cost indicators as separate descriptive characteristics. These measures are presented together to show how airline operating structures differ, without treating one measure as a direct explanation of another.

Figure 8. Turnaround time by airline. Data Source: Flightradar24; author-derived observations. Graph made by author.



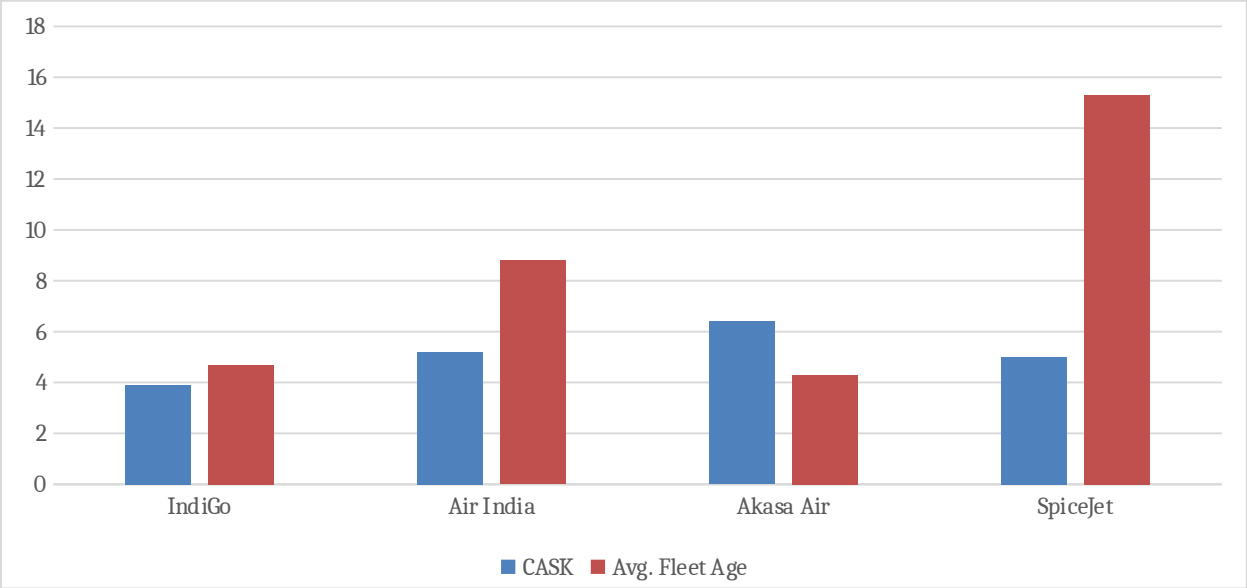
Turnaround observations were derived from flight-tracking records for a common Delhi–Mumbai service. The elapsed time between the recorded arrival of an aircraft and its subsequent departure was used as the operational observation. Because the manuscript did not identify an official, directly comparable turnaround series, this figure should be interpreted as a case-based descriptive comparison rather than a general system-wide measure.

Figure 9. Fleet age by airline. Data Source: Statbase, Aircraft Fleet Average Age by Airlines. Graph made by author.



Fleet structure provides additional context for the description of airline operations. The source material records differences in fleet composition among the carriers considered. IndiGo is described as operating a comparatively standardized fleet, whereas Air India's fleet includes a broader range of aircraft types associated with both domestic and international operations. SpiceJet and Akasa Air are also represented through their respective aircraft families in the supplied record.

Figure 10. Comparison of fleet age and CASK by airline. Data Source: sources identified in the data-provenance register. Graph made by author.

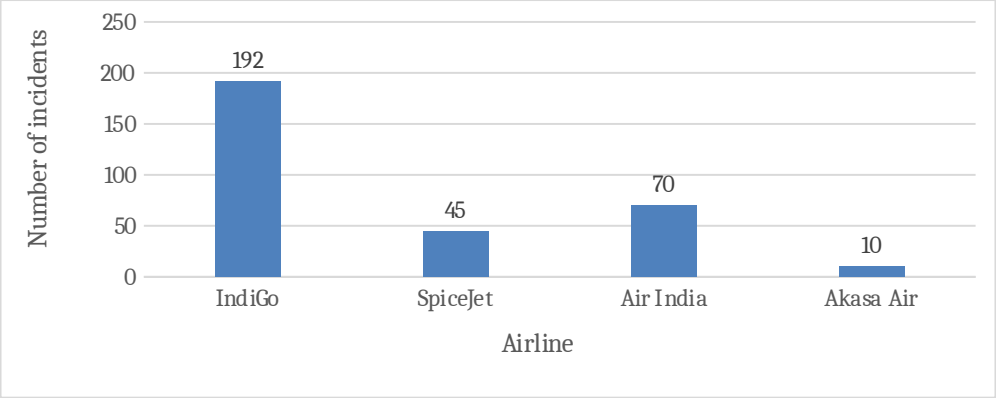


The retained comparison places fleet-age observations alongside CASK observations. It is presented as a descriptive comparison of two operating indicators. The figure does not, by itself, establish that fleet age causes changes in operating cost, and the revised paper does not constitute a causal test.

The operational record therefore describes several characteristics of the LCC model: fleet standardisation, aircraft utilisation, turnaround practices and cost discipline. These characteristics coexist with requirements for maintenance, staffing, scheduling and regulatory compliance.

5.4 Safety and Regulatory Context

Figure 11. Reported aviation incidents by airline. Data Source: Tiwari (2022), India Aviation Grievance Dataset, Kaggle. Graph made by author.



The incident dataset contains reported events related to the carriers that are represented in the source. Incident occurrences are not safety rates and are not comparable without taking into account exposure, fleet size, traffic volume, incident definitions and reporting. The record supplied also differentiates between routine incidents and serious accidents and highlights the need for regulatory oversight on the interpretation of aviation safety.

The provided 2025 observations also provide details of a significant disruption in operations, namely with IndiGo. The episode coincided with the new DGCA Flight Duty Time Limitation (FDTL) rules and was linked to the widespread cancellation of flights. The regulatory measures instituted were temporary exemptions, fare controls, greater monitoring and investigation. The episode is relevant to the descriptive account because it provides an example of the operation interdependency between crew planning, scheduling capacity and the regulatory requirements.

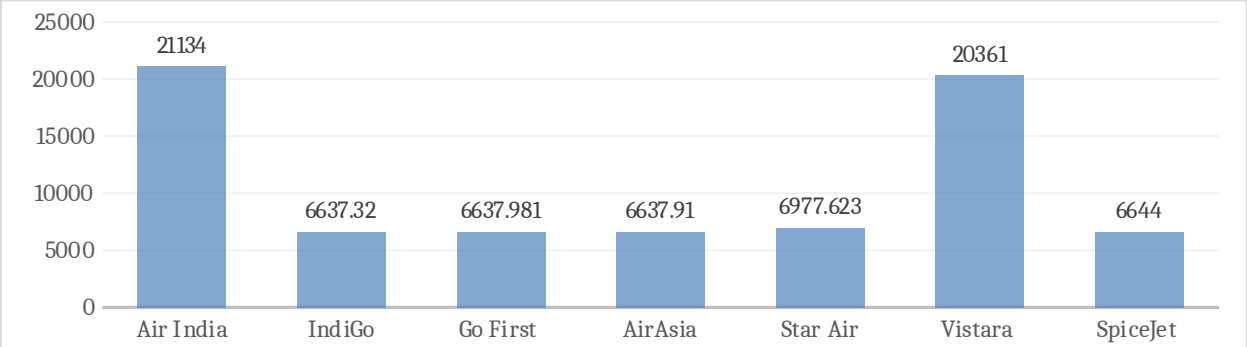
5.5 Fare Structure and Service Segmentation

The prices of fares provided in the supplied material are reported as observations of the prices of the base ticket across the various airlines. The LCC model typically includes a base charge for a ticket minus any optional services like checked baggage, seat selection, food, priority services and changes. Headline fares do not therefore reflect the total fare for each individual passenger; they are only a reflection of the price tag on the initial purchase of the ticket.

Dynamic pricing is a characteristic that has been mentioned in published research of the airline industry. The prices may be different depending on the availability of seats and the demand in the case of booking at different times. The literature, then, considers LCC pricing to be part of a more comprehensive pricing policy that becomes a low-price schedule only for a limited time. (Wittman & Belobaba, 2018; Wittman & Belobaba, 2019).

The fare figure and the separation between low cost and full-service fare structures are captured between the carriers in the dataset. The observations should be perceived as a description of the data set for that period—not necessarily as an average fare for all the routes and/or booking conditions.

Figure 12. Domestic fare observations by carrier, 2023. Data Source: Sahu (2024), Goibibo Flight Data, Kaggle. Graph made by author.

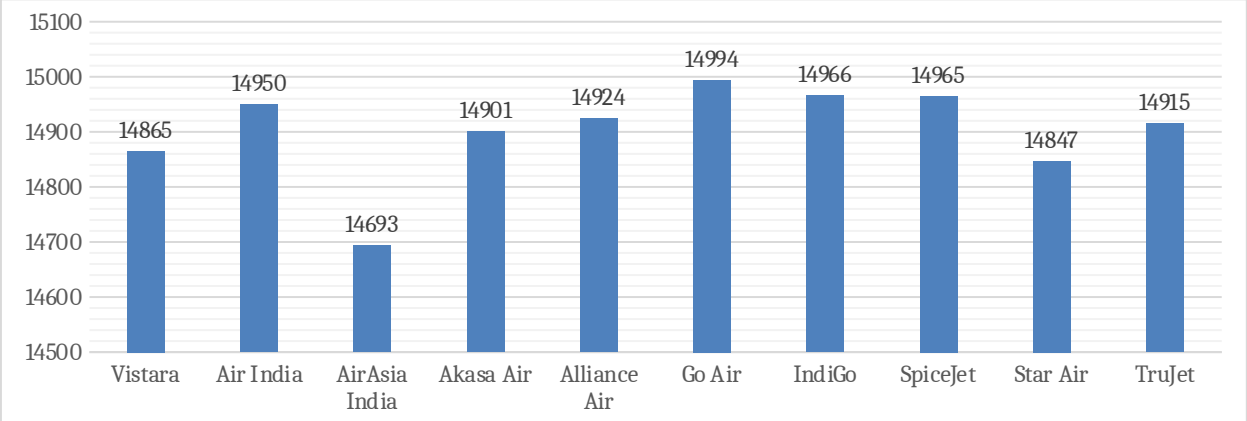


In addition to ancillary revenue, the LCC model depends on other revenue streams. Services can be bundled with the basic ticket and sold as add-on products. Services can be bundled and sold as add-on products and be separated from the basic ticket. This enables airlines to offer a lower far premium at the outset, and to make further profits via offering extra services to those passengers who choose to buy them.

The difference between the quoted rate and actual cash paid (after any optional services are added) thus influences fare comparison. Literature provided in the package points to transparency as key in understanding the distinction of airline pricing.

The next figure shows the number of flights over each of the carriers in the data set.

Figure 13. Flight volume by airline, 2019–2025. Data Source: Sahu (2024), Goibibo Flight Data, Kaggle. Graph made by author.



The flight-volume figure provides a further description of activity across LCC and full-service carriers. The continued presence of both business models indicates that the development of LCCs has occurred alongside, rather than simply in place of, full-service operations. The two models serve overlapping domestic markets while retaining differences in service structure, network strategy and passenger segmentation.

5.6 Cross-Domain Descriptive Synthesis

Overall, this is a descriptive interpretation of the evidence, with the LCC model serving as one of the key organisational aspects of today's domestic Indian aviation industry. It can be seen in a variety of areas: airline participation, development of networks, level of aircraft use, fare structure, and the organisation of optional services. Meanwhile, the evidence suggests that these domains are situated in a bigger scenario that is influenced by infrastructure, rules and regulations, economic factors, airline exits and mergers. The resultant image is then one of structural change, not one of the effect of one variable.

An extra qualification for safety and regulatory evidence. The number of incidents provided in the provided data set cannot be converted into direct airline safety rates since the number of incidents do not reflect exposure, traffic volume, reporting completeness or incident severity. Regulatory events (such as the 2025 IndiGo scheduling disruption that is described in the manuscript) are more effectively understood as case evidence in the context of an operation. They illustrate the significance of crew-duty regulations, capacity and compliance with regulations in a high utilisation airline system, but do not form a basis for a general safety rating of carriers.

The fare evidence also calls for something beyond just the low-price/high-price contrast. The base fare is just one part of an airline product, and the price of a ticket seen in a set of prices is conditioned by the circumstances of the ticket being recorded, due to dynamic pricing. The advertised fare is further broken down from the total travel package by ancillary services. The descriptive finding is that the carriers in the fares provided in the supplied fares data set are not all in the same position when it comes to pricing and service; it is not that all passengers on all LCC routes necessarily pay less than all the passengers using a full-service carrier.

These operational observations add to this description. Turnaround time, indicators for operating costs, and fleet age are different dimension of airline production. A short turnaround is the time between two movement cycles at an airport; fleet age is the age distribution of the aircraft in the fleet and CASK is an operating cost indicator based on capacity and distance. Not one of these indicators alone is sufficient to depict operational efficiency. Taken together, they describe the operating requirements for the various carriers to provide domestic service.

When considered together, the separate descriptive domains become more descriptive. The evidence of India's domestic market has witnessed frequent entry and exit and the connectivity evidence demonstrates the presence of larger metros as well as services to smaller metros. The records of passengers and airports give a fuller picture of a growing and evolving air-transport system for domestic travel. The LCC model is therefore embedded in a sector which has transformed in terms of its customers, infrastructure and its participants.

6. DISCUSSION

Language Arts: 6.1 Synthesis of the Descriptive Evidence

The evidence together indicates that the LCC is a part of a wider transformation in domestic aviation in India. The sector has been a business that has been going through entry, growth, exit and consolidation and LCCs have now joined the full-service carriers to become legitimate competitors. The retained market-share and carrier-participation figures illustrate this structural shift and the connectivity and

passenger numbers reflect the parallel development of domestic air-travel infrastructure and activity. Markets are characterized by a few large firms. The market is oligopolistic and connected.

Connectivity is also multi-dimensional. The airport and the passenger data provide information on the physical and traffic size of the domestic aviation market, whereas the descriptions of the carriers reveal services outside the largest metropolitan areas. The connectivity of smaller cities is influenced by the network decisions of the airlines, infrastructure of the airports and public programmes like UDAN. The evidence in the descriptions thus documents the LCC model as part of a more general process in which domestic air travel was made available at a higher number of sites.

The aviation industry is an example of the evidence of a change in market structure - number and kind of airlines. Each of the existing airlines and new airlines (or market exits) are added to the chosen year's airline table. The diversity of the makeup is crucial as it is not as if the LCC model had evolved in isolation—the evolution of the LCCs has occurred in parallel with the development of full-service carriers, airline groups and regional connectivity programs. Hence, it would not be possible to discuss the market based on the history of any individual airline. Ensuring the efficacy of the system of operations, fares and regulation.

These commercial and operational practices are institutionalized in the form of regulation. DGCA regulates the Indian aviation system in terms of airworthiness, flight operations, crew licensing and related aspects. The operation of the system cannot be optimised without taking into consideration the regulatory requirements and the planning capacity, as demonstrated by the example of disruptions in 2025 in the present paper. The event does not claim to be a general relationship between the LCC model and safety outcomes; instead, it is presented as a case, which is described in the descriptive record.

One of the major advantages of the LCC model is the distinction made between the basic ticket and any services that might be added to it, as illustrated in the fare observations. This can make the initial outlay significantly vary from the cost of a full-service offering. The other component of dynamic pricing is that a fare is time and demand sensitive. Therefore, if a traveller demanded a descriptive comparison, it should be possible to maintain a difference between their base fare recorded and the complete set of services they used.

The evidence indicates that cost-oriented airline organisation involves a series of related practices and is not one technological innovation. Fleet commonality, aircraft usage, ground handling, scheduling etc. all impact the use of an airline's resources. The turnaround comparison is also helpful as a tangible example of the process: it's built on the specifics of each route, and shouldn't be used as an indicator of the overall performance of each airline.

7. CONCLUSION

The descriptive record demonstrates that low-cost carriers have now been an integral part of a domestic aviation system in India. There is a familiar model that can be recognised for their development which is based on simplified service structures, fleet strategies, aircraft utilisation, turnaround practice and differentiated pricing. Meanwhile, full-service carriers, regional operations and evolving airline ownership and involvement continue to remain part of the domestic market.

The data further reveals that development of airlines is closely linked to the regulatory environment. Regardless of business model, safety oversight, crew duty requirements, maintenance arrangements and

planning of operations are essential to airline operations. Large scale interactions between scheduling and regulatory requirements are illustrated by the disruption of the IndiGo in 2025 that was supplied.

The LCC model is not limited to a low ticket price, as connectivity and fare data show. Its service structure typically includes extra services as add-on to the base service, and airline networks and airport availability dictate the actual availability of air travel. The simultaneous presence of two types of carriers (LCC and full-service carriers) thus suggests differentiation of products and operating strategies in a domestic market.

In general, this paper describes the structural, functioning and commercial features of LCC development in India. It is not the intention of this study to attribute the observed changes to LCC entry exclusively. Rather, it documents the evolution of the LCCs in the context of liberalisation, infrastructure growth, airline exit and entry, consolidation, economic upheaval and ongoing regulation.

8. FUTURE RESEARCH

Other studies could build on this descriptive record by looking at changes along the route, environmental performance development, the impact of market consolidation, the evolution of newer LCCs and aircraft technology evolution. More detailed longitudinal data could even differentiate changes due to individual carriers, routes, airport infrastructure and broader economic conditions.

9. LIMITATIONS

The main drawbacks are the uniformity and the comprehensiveness of the secondary record. The classification of air lines have also evolved over time, some airlines have ceased operations and the description or composition of specific DGCA series may vary between reporting periods. All events are not necessarily captured consistently in all incidents, foreign-carrier activity can affect the domestic aviation environment and fuel price changes or significant disruptions can affect the domestic aviation environment. A few figures were also created or calculated by the authors since a series of official figures was not found in the provided material that is comparable. Thus, it is important to read the paper as a description of what evidence is available, and not as a causal analysis of the LCC business model.

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